Supplementary Material
Observer-Based Hybrid Position Holding for an Electrohydraulic Agricultural Hitch

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S1. Historical model and corrected interpretation

The historical pressure-difference surrogate and V2 are separate models. The old surrogate has a virtual command-to-pressure relaxation, a retained-loss multiplier and a virtual accumulator/recharge proxy. Its paired pressure outputs are reconstructed rather than separately conserved. The implemented pressure loss is $\mathcal{L} = \kappa[1 + g_T \max(T - 25, 0)] \max(d, 0)/A$. Dimensional consistency requires $[\kappa] = \mathrm{m}^2\,\mathrm{s}^{-1}$ so that $[\mathcal{L}] = \mathrm{Pa}\,\mathrm{s}^{-1}$. The old physical leakage-flow interpretation and $\mathrm{m}^3\ \mathrm{s}^{-1}\ \mathrm{Pa}^{-1}$ label are withdrawn. Old file identifiers are retained only to reproduce historical computations. The corrected coefficient-proxy inversion and a rate-dependent bound appear in Section 3.5. They do not identify physical leakage in V2.

The archived raw 0.10 MPa pressure-noise group (seeds 2060–2079) has 500 mm maximum steady error in every path, with mean steady RMSE 486.56 mm. A small reported downward drop of 0.033 mm in that group is not successful holding: the actuator travels to the upper stroke boundary. This illustrates why displacement drop alone is an insufficient acceptance metric. The V2 replay at the same absolute pressure-noise SD has 20/20 passes and no stroke hit, but its observer, reference and draw sequence also differ. This is a comparison of implementations, not a controlled attribution to chamber architecture alone.

S2. Measurement interface and original selection budget

The full observer update, pressure command, saturation, integrators, switching order, hysteresis and timers are in Sections 3.2–3.3. Position correction is 0.08, velocity correction is $0.0005/h_c$, pressure filtering is 0.18 and effective-load smoothing is 0.005 per controller sample. Initial position and pressures equal the initial plant values, velocity is zero, and load starts at 9000 N. This initialization is a virtual-study assumption; it is not evidence of physical observer convergence. The effective load is clipped to [3000,20000] N in the controller. Nominal model coefficients remain fixed when true mechanical parameters are varied.

The 27 development candidates cross dwell $\{4, 8, 12\}$ s, support-error threshold $\{2.5, 4, 6\}$ mm and support-pressure threshold $\{0.18, 0.30, 0.45\}$ MPa, evaluated for 300 s on seeds 4100–4104. All pass five development paths. Ranking is lexicographic: maximum pass count, then minimum mean oil volume, mean mode changes and mean drop. Equal ties preserve enumeration order. No revision result changes the selected settings. The complete development table follows; quantities are virtual inputs or outputs, not hardware calibration.

Table 1: Original development grid; every row passes 5/5.

Dwell (s)	Error (mm)	Deficit (MPa)	Oil (mL)	Switches	Drop (mm)
4	2.5	0.18	21.27	5.0	0.996
4	2.5	0.30	15.50	3.4	1.222
4	2.5	0.45	15.50	3.4	1.222
4	4.0	0.18	21.27	5.0	0.996
4	4.0	0.30	0.00	3.0	1.773
4	4.0	0.45	0.00	3.0	1.773
4	6.0	0.18	21.27	5.0	0.996
4	6.0	0.30	0.00	3.0	1.773
4	6.0	0.45	0.00	3.0	1.773
8	2.5	0.18	25.44	5.0	0.845
8	2.5	0.30	17.98	3.0	1.230
8	2.5	0.45	17.98	3.0	1.230
8	4.0	0.18	25.44	5.0	0.845
8	4.0	0.30	0.00	3.0	1.754
8	4.0	0.45	0.00	3.0	1.754
8	6.0	0.18	25.44	5.0	0.845
8	6.0	0.30	0.00	3.0	1.754
8	6.0	0.45	0.00	3.0	1.754
12	2.5	0.18	30.46	4.6	0.861
12	2.5	0.30	20.82	3.0	1.188
12	2.5	0.45	20.82	3.0	1.188
12	4.0	0.18	30.46	4.6	0.861
12	4.0	0.30	0.00	3.0	1.803
12	4.0	0.45	0.00	3.0	1.803
12	6.0	0.18	30.46	4.6	0.861
12	6.0	0.30	0.00	3.0	1.803
12	6.0	0.45	0.00	3.0	1.803

S3. Original frozen statistics and uncertainty

The submitted 20 paths use seeds 4120–4139, disjoint from development. Mean confidence intervals use $\bar{z} \pm t_{0.975,n-1} s/\sqrt{n}$, where s is sample SD. These are intervals for seeded model-noise means, not hardware population intervals or pass-probability guarantees. No seed is removed for failure. Drop statistics require hold entry; every submitted path enters holding. The post-transition maximum error and pass screens include failed paths.

Chain	Metric	Mean	SD	95% CI
PID–lock	Drop (mm)	16.041	0.065	16.010–16.071
PID–lock	RMSE (mm)	8.232	0.115	8.178–8.286
PID–lock	Max. error (mm)	15.138	0.132	15.076–15.200
PID–lock	Mode changes	1.000	0.000	1.000–1.000
PID–lock	Oil (mL)	0.000	0.000	0.000–0.000
SMC–lock	Drop (mm)	15.995	0.055	15.969–16.021
SMC–lock	RMSE (mm)	9.174	0.175	9.092–9.256
SMC–lock	Max. error (mm)	16.227	0.199	16.134–16.320
SMC–lock	Mode changes	1.000	0.000	1.000–1.000

Chain	Metric	Mean	SD	95% CI
SMC–lock	Oil (mL)	0.000	0.000	0.000–0.000
PID–accumulator	Drop (mm)	3.441	0.249	3.325–3.558
PID–accumulator	RMSE (mm)	0.246	0.004	0.244–0.247
PID–accumulator	Max. error (mm)	6.960	0.219	6.858–7.063
PID–accumulator	Mode changes	26.800	5.872	24.052–29.548
PID–accumulator	Oil (mL)	25.538	1.200	24.977–26.100
Observer hybrid	Drop (mm)	1.849	0.156	1.776–1.922
Observer hybrid	RMSE (mm)	1.222	0.026	1.210–1.234
Observer hybrid	Max. error (mm)	8.435	0.219	8.332–8.537
Observer hybrid	Mode changes	13.100	0.447	12.891–13.309
Observer hybrid	Oil (mL)	1.019	4.559	-1.114–3.153

S4. Integration refinement and local boundedness checks

Sensing, observer updates and commanded inputs remain on the 10 ms grid. Only the plant is substepped at 10, 5, 2.5 or 1 ms; the measurement draws are identical on common controller samples. Pressure updates use the old position/velocity and the updated spool, then the mechanical update uses the new pressures, and position uses the new velocity. This sequential explicit scheme and its pressure/stroke projections are retained. It is not an adaptive error-controlled solver. Pressure discrepancies below aggregate both chambers on all 10 ms sample endpoints against the 1 ms run. Maxima may include a switching-time displacement and are not an estimate of local truncation order.

Table 3: Full-horizon 10 ms versus 1 ms pressure and drop differences.

Seed	Pressure RMS (MPa)	Max. difference (MPa)	Drop difference (mm)
4300	0.0638	0.1510	0.1990
4301	0.0642	0.1657	0.1120
4302	0.0640	0.1583	0.2394
4303	0.0633	0.1543	-0.0326
4304	0.0633	0.1522	0.2357

The event-clipped RK4 audit keeps controller, observer and measurement updates on the 10 ms grid and changes only the virtual-plant update. Five predeclared seeds were run at 1.0 and 0.5 ms; all paths passed. The mean hold drops were 1.628 and 1.585 mm, and the mean worst post-transition errors were 8.503 and 8.502 mm. The mean absolute per-seed difference in worst error was 0.0023 mm and the mean absolute hold-drop difference was 0.046 mm. These results support fine-step numerical convergence of the refined arm, while switching and friction remain sources of hold-drop sensitivity.

Table 4: Event-clipped RK4 fine-step convergence audit.

Plant step (ms)	Pass	Mean drop (mm)	Drop SD (mm)	Mean worst error (mm)	Worst error (mm)
1.0	5/5	1.628	0.058	8.503	8.747
0.5	5/5	1.585	0.059	8.502	8.747

The projected-model velocity bound is independent of mode but depends on imposed pres-

sure/stroke limits. It is deliberately too loose for a 10 mm guarantee. For the unprojected local calculation at $\bar{x} = 0.35$ m, nominal $C_l = 5 \times 10^{-14}$ m^3 s^{-1} Pa^{-1} and 25 °C, the frozen-volume matrix in Section 3.4 has eigenvalues approximately -6.23×10^{-6} and $-0.918 \pm 73.08i$ s^{-1}. The slow pole exposes weak leakage-driven restoration rather than precise locking. The Lyapunov argument is in fixed consistent coordinates, not a physical-energy interpretation of a mixed-unit Euclidean norm. The result concerns bounded forcing and fixed volumes, with no claim that the nonlinear trajectory remains in its local neighborhood.

S5. Noise activation, storage and historical alignment

Independent Gaussian pressure noise is added before the V2 internal pressure filter; no new external low-pass filter is inserted. Position noise stays at 1.5 mm SD in the pressure-only sweep. The separate joint-noise case doubles both SDs. Dwell limits mode changes but does not prevent persistent positive support commands during a latched support mode. Pressure/position fluctuations change pressure balance and gate crossings, explaining the large change in oil volume from a near-zero nominal denominator. Gas pressure, cumulative discharge, active time and mode occupancy are recorded; energy consumption and valve wear are not inferred from these proxies.

Table 5: Five-path noise-group activity and storage extrema.

Noise input	Pass	Active time (s)	Mean oil (mL)	Min. gas (MPa)
Nominal	5/5	0.00	0.00	14.000
Both SDs doubled	5/5	6.23	39.92	13.553
Pressure SD 0.10 MPa	5/5	18.51	98.76	13.325
Pressure SD 0.20 MPa	5/5	177.44	298.04	12.204
Pressure SD 0.50 MPa	5/5	450.62	447.11	11.497
Pressure SD 1.00 MPa	2/5	479.06	454.13	11.466

The ten original jointly doubled-noise paths were replayed with diagnostic logging before the timestamp-schema correction; all original fields match exactly in that frozen replay. Mean support-command active time is 10.391 s (SD 5.196 s), mean oil volume 61.982 mL (SD 28.408 mL), and the minimum gas pressure across the group is 13.337 MPa. Every path passes the original screen and has no stroke hit. Figure 10 of the main article shows the first original stress seed 4160, against a same-seed nominal replay, so that the original discharge increase has a direct trajectory diagnosis rather than being explained only through a new seed set.

The aligned 20-path historical-noise replay gives mean drop 1.860 mm (95% interval 1.774–1.946), worst error 8.723 mm, and mean oil 105.36 mL. Its larger discharge is compatible with passing the position screen. Seeds and SD match the old experiment, but the stochastic samples, plant, reference and internal observer do not. An architecture-only noise-robustness causal claim is therefore withheld.

S6. Parameter provenance and revision overrides

The original 45-row parameter manifest is supplied without rewriting historical coverage labels. Every number is a researcher-defined theoretical input. The new ledger records additional true-plant overrides, integration substeps, observer-removal arms and noise scales separately. Hydraulic area, dead volume, effective bulk modulus, flow area and dead zone have not been independently identified. A declared range in the submitted table did not by itself document a performed

test. The new mechanical runs vary one true coefficient at a time and keep controller/observer inputs nominal; they cannot establish robustness over all simultaneous combinations.

S7. Reproduction and integrity

The reproduction ZIP contains current code, original JSON evidence, revision protocols/results, parameter manifest and scripts. A frozen submitted source/code copy is kept under historical submission evidence. The control defaults reproduce all original numerical fields on one original path of each of the four chains; the corrected hold-start timestamp is the only changed legacy field (S10). This is a regression check, not a complete numerical validation. The original 20-path JSON values are retained rather than overwritten. Run the scripts below from the reproduction root with Python 3 and NumPy, SciPy and Matplotlib. Their protocol files specify seeds before computation. The mode-scan script is historical and need not be rerun to reproduce the fixed revision evidence.

```
python3 tools/run_paper1_major_revision.py
python3 tools/extend_paper1_revision_audits.py
python3 tools/refine_paper1_comparators.py
python3 tools/replay_paper1_original_noise.py
python3 tools/plot_paper1_major_revision.py
```

The first script runs 177 primary revision simulations; the historical-noise alignment adds 20, refined comparator checks add 15, and original-noise diagnostics add 11. These are seeded theoretical replays, not 223 independent physical experiments; families deliberately reuse paired seeds. The first primary revision protocol is supplemented by the explicit mechanical-mismatch ledger. Results and source hashes are listed in the package manifest. Main article figures use predeclared revision seed 4300 or the first original noise seed 4160, not best-path selections. Public GitHub/Zenodo deposition of this revision remains pending and is not asserted as complete.

S8. Paired RK4 comparators and joint mechanical vertices

The strengthening protocol fixes all 70 rows before computation: 30 nominal comparator rows and 40 joint-vertex rows. Ten nominal hybrid rows are reused from the earlier RK4 audit, so 60 new trajectories are computed. Each path lasts 1800 s, with controller/observer/noise updates at 10 ms. The two PID arms use exactly the same five seeds and plant integrator as the hybrid; the matched arm changes only the declared latch and discharge-gating semantics. The joint screen crosses all eight $\{0.7, 1.3\}^3$ mass/damping/stiffness vertices with five seeds, a 1 ms RK4 plant step and fixed nominal control coefficients. Source hashes, per-task checkpoints and all raw outputs accompany the protocol. These vertices do not certify the full box or an uncertainty probability distribution. Endpoint projections do not locate event roots or establish a fourth-order rate for this nonsmooth system.

Table 6: Paired RK4 comparison on five common seeds per row, with frozen gains and 10 ms sensing/control. The matched PID arm adds an 8 s latch and discharge gating. RMSE is post-transition tracking error; oil is accumulator discharge.

Chain	Step (ms)	Pass	Mean drop (mm)	Mean RMSE (mm)	Mean oil (mL)	Mean changes
Observer hybrid	1.0	5/5	1.628	1.178	0.00	13.0
PID–accumulator	1.0	5/5	3.002	0.249	29.31	40.6
PID + matched latch	1.0	5/5	3.028	0.249	27.94	3.0
Observer hybrid	0.5	5/5	1.585	1.192	0.00	13.0
PID–accumulator	0.5	5/5	2.943	0.251	29.91	42.6
PID + matched latch	0.5	5/5	2.953	0.251	28.50	3.0

Table 7: Joint mechanical corner screen with the hybrid, five paired seeds per vertex and 1 ms RK4 integration. All controller/observer coefficients remain nominal. Ratios refer to true plant mass, damping and stiffness. Worst values include every path.

m/m_0	c/c_0	k/k_0	Pass	Worst drop (mm)	Worst error (mm)	Mean oil (mL)
1.3	1.3	1.3	5/5	1.595	9.155	0.00
1.3	1.3	0.7	5/5	1.756	8.352	0.00
1.3	0.7	1.3	5/5	1.583	9.023	0.00
1.3	0.7	0.7	5/5	1.692	8.284	0.00
0.7	1.3	1.3	5/5	1.746	9.153	0.00
0.7	1.3	0.7	5/5	2.009	8.351	0.00
0.7	0.7	1.3	5/5	1.750	9.177	0.00
0.7	0.7	0.7	5/5	1.965	8.372	0.00

Table 8: Paired PID-minus-hybrid differences in hold drop and RMSE with Student-t 95% intervals, five paired seeds per row. Intervals concern seeded noise at the declared plant, not uncertainty in model structure or hardware performance.

PID arm	Step (ms)	Drop difference (mm), 95% CI	RMSE difference (mm), 95% CI
Original PID	1.0	1.374 [1.149, 1.599]	-0.929 [-0.962, -0.896]
Matched PID	1.0	1.400 [1.147, 1.652]	-0.929 [-0.962, -0.896]
Original PID	0.5	1.358 [1.091, 1.625]	-0.941 [-0.966, -0.917]
Matched PID	0.5	1.368 [1.106, 1.630]	-0.941 [-0.965, -0.917]

```
python3 tools/run_paper1_rk4_convergence_audit.py
python3 tools/run_paper1_strengthening_audit.py
```

S9. Physical-system and combined-stress audit

Figure 1 records the physical-system abstraction used by the V2 equations. The piston coordinate is the simulated displacement; an implement displacement would require an external linkage map and is not calibrated here. The observer uses noisy position and chamber-pressure measurements at 10 ms. The symmetric equal-area assumption is retained as a declared virtual-model boundary.

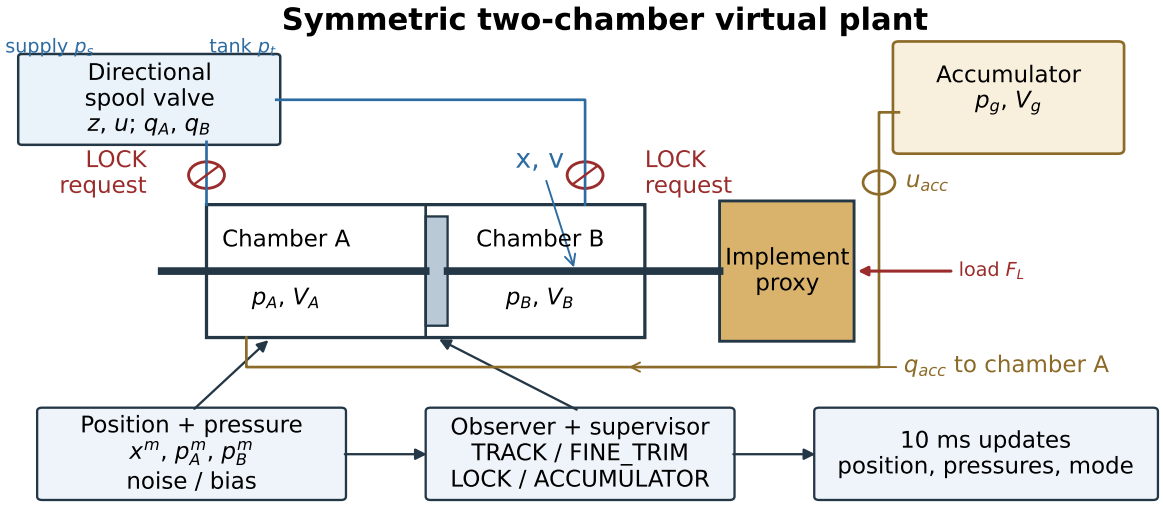


Figure 1: Hydraulic abstraction with conceptual LOCK requests; see the main article for their numerical implementation.

The combined stress protocol was declared before execution and uses load changes at 0, 60, 300, 600, 900 and 1200 s, five seeds (4400–4404), the 1800 s reference and the unchanged 10 ms observer/controller. The nominal arm uses the declared leakage coefficient. The second arm multiplies it by five; the third uses the same fivefold coefficient with the accumulator valve disabled. Each arm contains five traces and no case is selected after observing its result.

Table 9: Combined load/leakage stress audit.

Arm	Entry	Pass	Mean drop (mm)	Worst drop (mm)	Worst error (mm)
Nominal leakage, accumulator enabled	5/5	5/5	2.56	2.81	8.62
5x leakage, accumulator enabled	5/5	5/5	2.68	3.77	8.63
5x leakage, accumulator disabled	5/5	5/5	2.68	3.77	8.63

The enabled fivefold arm reached a minimum accumulator pressure of 13.59 MPa and used 36.02 mL mean oil on these traces; the nominal arm used 38.13 mL. The disabled arm also passed 5/5, so this protocol does not show that the accumulator is necessary. It does show the requested ablation result and makes the limitation explicit: stronger storage-demanding disturbances, recharge dynamics and calibrated leakage remain untested.

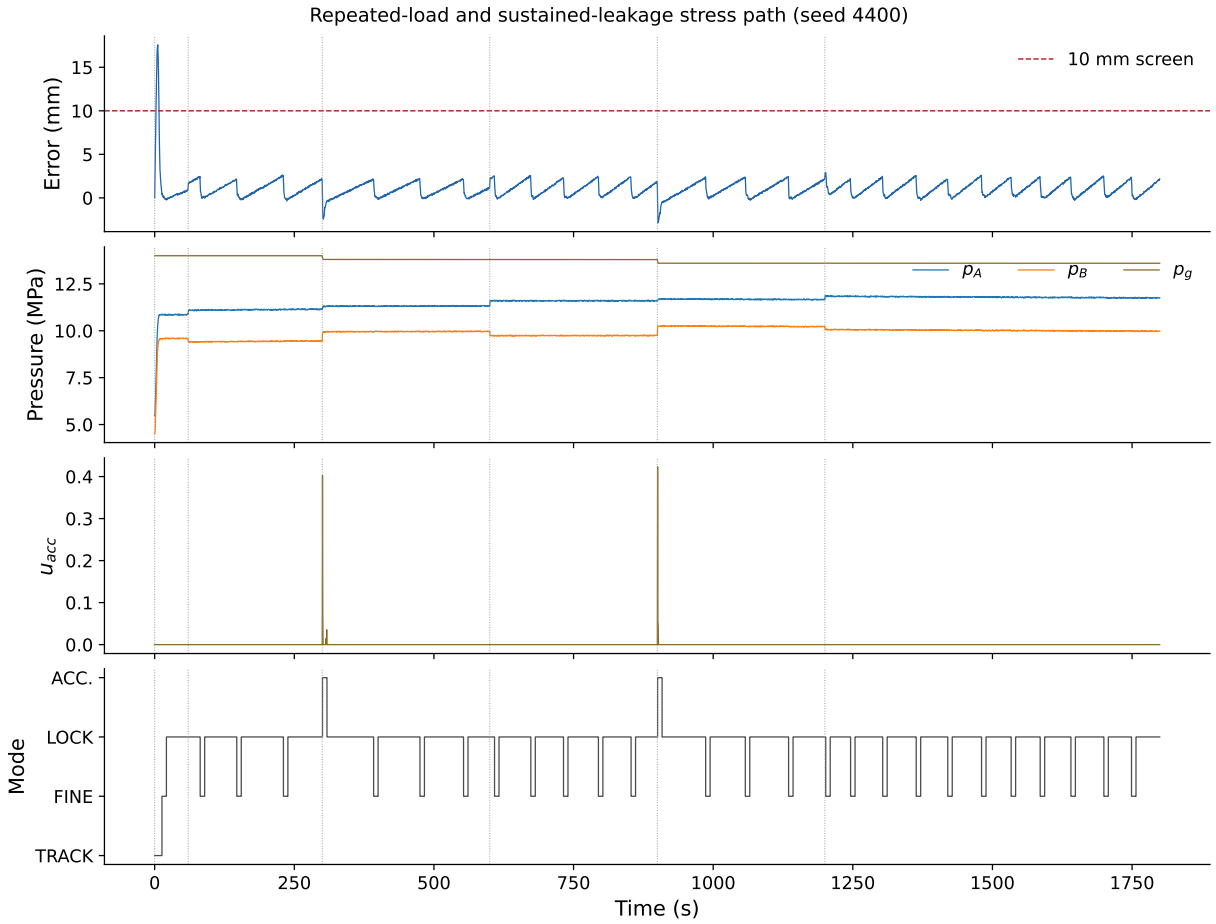


Figure 2: Representative combined-stress trace, seed 4400.

The friction-mismatch extension uses the same five seeds, repeated-load schedule and fivefold virtual leakage. In the 0.7× arm the true-plant viscous, Coulomb and Stribeck terms are all reduced together; in the 1.3× arm they are all increased together. The observer and controller retain nominal force-balance coefficients; neither explicitly compensates Coulomb/Stribeck friction, whose effects can enter the effective-load estimate. Both arms enter and pass on 5/5 paths, with worst errors of 8.63 and 8.65 mm and worst drops of 2.88 and 2.99 mm, respectively. S10 adds independent friction amplitudes/velocity scales and interacting parameters. Neither finite screen identifies a physical friction law or guarantees the full region.

Table 10: Friction-mismatch extension to the combined-stress audit.

True-plant friction	Pass	Mean drop (mm)	Worst drop (mm)	Worst error (mm)	Mean oil (mL)
0.7× nominal	5/5	2.46	2.88	8.63	43.14
1.3× nominal	5/5	2.47	2.99	8.65	37.09

S10. Independent tuning and uncertainty audits

Additional independent risk-closure protocol. The frozen submitted gains remain distinct from a new equal-budget development comparison. Each policy receives nine candidates. Tracking multipliers are $g_t \in \{0.7, 1, 1.3\}$; accumulator multipliers are $g_a \in \{0.5, 1, 1.5\}$. The multiplier g_t scales the 650 and 90 feedback gains and the PID integral gain 20, but leaves reference feedforward

unchanged; g_a scales the complete accumulator command before saturation. Both policies use the same ± 0.10 main-valve authority, 8 s dwell and mode-gated discharge. Mode-request thresholds remain policy-specific. Thus authority, timing and search effort are matched without making the policies identical. Three development seeds (4600–4602), 300 s paths and 1 ms RK4 are used per candidate. The development load schedule is $(0, 9), (60, 14), (100, 10), (160, 16), (220, 11), (260, 15)$ kN with fivefold leakage. Candidates are ranked by failure count, then mean $J = \Delta_h/10 + \mathrm{RMSE}/10 + V_{acc}/500 + N_{sw}/225$, using mm and mL. These fixed engineering normalizations express one preference, not universal utility weights. All candidates and their four-objective nondominated subset are retained. Selection is recorded before the independent 1800 s evaluation on seeds 4700–4709, using the full repeated-load schedule at both 1 and 0.5 ms. No validation outcome changes selection. Both policies share each seed's measurement sequence and initial plant state. This finite grid is not a global optimization or a comparison against all robust-control methods.

A separate convergence screen repeats nominal leakage, fivefold leakage, accumulator-disabled and linked $0.7/1.3\times$ friction arms at 1 and 0.5 ms on seeds 4400–4402, without retuning the submitted controller. A 24-path stratified interior screen uses new seeds 4800–4823 and independently permuted bins for 12 simultaneous inputs: mass, viscous damping, stiffness, Coulomb friction, Stribeck amplitude and velocity scale, bulk modulus, spool lag, leakage multiplier, accumulator flow coefficient, pressure-noise amplitude and position bias. The intervals and all sampled inputs are supplied in S10. These are virtual-design intervals, not calibrated distributions. The controller retains the original gains and nominal force-balance coefficients. It contains no explicit Coulomb/Stribeck compensation; these unmodeled terms can enter its effective-load estimate. Independent friction magnitudes and velocity scales test more than a linked scaling, but the functional friction law itself remains fixed.

The current metric schema records the confirmation sample and first post-update endpoint separately. Historical numerical outputs are unchanged; the old start label is retained as legacy metadata. A four-chain regression on seed 4120 reproduces every historical output field except the corrected start timestamp. The confirmation gate counts 100 samples; it is not reinterpreted as a continuous-time crossing. The new metric records the start position, minimum hold position and sample count so that the reported drop is directly checkable.

Scope of mathematical statements. The sequential projected-state recurrence uses the actual friction bound and integration step of each plant. It bounds numerical states without guaranteeing millimeter accuracy. Positive parameters satisfy the frozen-volume Routh test pointwise, but the corresponding Lyapunov matrix depends on those parameters. Neither argument proves stability of the nonlinear switched unprojected plant. The RK4 and uncertainty results are simulation evidence, not replacements for such a proof.

Declared simultaneous intervals. Mass 840–1560 kg; viscous coefficient 1540–2860 N s m^{-1}; stiffness 12600–23400 N m^{-1}; Coulomb friction 595–1105 N; Stribeck amplitude 245–455 N; Stribeck velocity 0.0028–0.0052 m s^{-1}; bulk modulus 0.14–0.26 GPa; spool lag 35–65 ms; leakage multiplier 1–10; accumulator-flow multiplier 0.5–1.5; pressure-noise multiplier 1–4; position bias -2 to $+2$ mm. Seed 48191 fixes stratification and independent bin permutations. All 24 input vectors are in protocol.json and individual checkpoints. The original submitted hybrid is used; these data do not tune a new robust controller.

Table 11: Complete development grid. Each candidate has three 300 s development paths. The nondominated flag uses drop, RMSE, oil and switching jointly.

Policy	ID	g_t/g_a	Fail	Mean J	Drop	RMSE	Nondom.
Hybrid	0	0.7/0.5	3	0.4695	2.478	1.411	No
Hybrid	1	0.7/1.0	3	0.5016	2.493	1.420	No
Hybrid	2	0.7/1.5	3	0.5234	2.479	1.411	No
Hybrid	3	1.0/0.5	0	0.4295	2.254	1.200	Yes
Hybrid	4	1.0/1.0	0	0.4577	2.276	1.204	No
Hybrid	5	1.0/1.5	0	0.4823	2.307	1.211	No
Hybrid	6	1.3/0.5	0	0.4521	2.471	1.127	Yes
Hybrid	7	1.3/1.0	0	0.4880	2.489	1.108	Yes
Hybrid	8	1.3/1.5	0	0.5209	2.514	1.165	No
PID	0	0.7/0.5	2	0.7174	3.990	1.174	No
PID	1	0.7/1.0	2	0.7998	3.870	1.151	No
PID	2	0.7/1.5	2	0.8504	3.799	1.136	No
PID	3	1.0/0.5	0	0.6106	3.843	0.891	No
PID	4	1.0/1.0	0	0.6713	3.748	0.874	No
PID	5	1.0/1.5	0	0.7160	3.678	0.864	No
PID	6	1.3/0.5	0	0.5476	3.666	0.748	Yes
PID	7	1.3/1.0	0	0.5968	3.606	0.738	Yes
PID	8	1.3/1.5	0	0.6346	3.560	0.729	Yes

Table 12: Paired matched-PID minus hybrid differences, independent ten-seed test. Student-t intervals summarize seeded model noise; they do not represent parameter or physical uncertainty.

Step (ms)	Quantity	Mean difference	95% interval
1.0	Drop (mm)	1.4651	[1.2326, 1.6976]
1.0	RMSE (mm)	-0.9519	[-0.9633, -0.9405]
1.0	Oil (mL)	10.7885	[7.0132, 14.5638]
1.0	Changes	-47.5000	[-48.1079, -46.8921]
0.5	Drop (mm)	1.4470	[1.2021, 1.6920]
0.5	RMSE (mm)	-0.9594	[-0.9719, -0.9469]
0.5	Oil (mL)	11.4952	[6.5113, 16.4791]
0.5	Changes	-47.2000	[-47.8574, -46.5426]

Table 13: Every joint interior sample, submitted hybrid at 1 ms RK4. Full input vectors accompany each checkpoint.

Seed	Entry	Drop (mm)	Max error (mm)	Relief	Stroke	Screen
4800	Yes	2.640	9.928	0	0	Pass
4801	Yes	2.684	6.628	0	0	Pass
4802	Yes	3.265	9.232	0	0	Pass
4803	Yes	2.470	8.661	0	0	Pass
4804	Yes	2.389	6.468	0	0	Pass
4805	Yes	1.899	8.465	0	0	Pass
4806	Yes	2.379	10.155	0	0	Fail
4807	Yes	2.054	6.932	0	0	Pass
4808	Yes	1.585	9.288	0	0	Pass
4809	Yes	2.396	7.314	0	0	Pass
4810	Yes	2.819	9.279	0	0	Pass
4811	Yes	2.224	8.047	0	0	Pass
4812	Yes	2.179	8.056	0	0	Pass
4813	Yes	2.058	7.372	0	0	Pass
4814	Yes	3.659	7.839	0	0	Pass
4815	Yes	2.208	6.563	0	0	Pass
4816	Yes	1.748	9.395	0	0	Pass
4817	Yes	2.423	5.735	0	0	Pass
4818	Yes	2.479	9.330	0	0	Pass
4819	Yes	2.966	9.587	0	0	Pass
4820	Yes	1.244	5.993	0	0	Pass
4821	Yes	2.056	10.004	0	0	Fail
4822	Yes	2.598	9.724	0	0	Pass
4823	Yes	2.734	7.555	0	0	Pass

```
python3 tools/replay_paper1_frozen_risk_closure.py --workers 4
python3 tools/report_paper1_risk_closure.py
```

S11. Verified active comparison and new joint design

Active nonlinear comparison and verified execution. The active sliding-mode (SMC) chain uses $1200\tanh[(18e + e_v)/0.006]$ N with the common observer, pressure PI, nominal force-balance coefficients and ± 0.10 holding authority, without the PID position integral. The main valve remains active during holding. The support threshold is $e > 2$ mm. Its accumulator command is

$$u_{acc} = \mathrm{clip}_{[0,1]}[80 g_a \max(e - 0.001, 0)].$$

Discharge additionally requires the latched ACCUMULATOR label. Both active baselines explicitly receive the same 8 s latch argument; their initial FINE_TRIM label is retained until the latch expires. The hybrid has its stated policy-specific hysteresis and the same 8 s minimum dwell. The control interfaces and budgets are matched, while the decision rules remain different. This is an executable boundary-layer comparator informed by the sliding-mode family (see the main-article sliding-mode references); it does not reproduce a published adaptive design or prove reaching of the full saturated loop.

A final execution audit found that an earlier SMC runner supplied zero rather than 8 s to the latch argument, also bypassing mode-gated discharge. Its records remain archived as an unlatched diagnostic and are excluded from matched-baseline claims. A separately frozen correction repeats all nine SMC candidates on the same development seeds 4600–4602, 300 s schedule, 1 ms RK4 and

unchanged scalar objective. Selection uses only those corrected development outcomes. Corrected SMC replays of common nominal seeds 4700–4709 at 1 and 0.5 ms, known vectors 4800–4823 and previously examined vectors 4900–4923 are follow-up checks, not fresh holdouts. Changes relative to the unlatched arm combine execution correction with reselection; they do not isolate a causal effect of dwell alone.

Separate accuracy preference and new joint verification. The scalar-selected hybrid and PID remain frozen at $(g_t, g_a) = (1, 0.5)$ and $(1.3, 0.5)$. The margin hybrid remains the original development-grid choice $(1.3, 0.5)$: require no development failure and worst development error ≤ 8 mm, then minimize the same J and candidate ID. Its 2 mm reserve is a heuristic for the stated bias amplitude, not a total observer-error bound. No result from a joint design changes these choices. Corrected SMC selection applies the same nine-candidate rule, and its eligibility under the 8 mm constraint is reported separately. Development-only weight profiles remain sensitivity checks rather than test-based tuning.

Before corrected development, random seed 51191 freezes 24 new stratified vectors over the same twelve intervals, with noise seeds 5100–5123. All four policy instances face identical true-plant overrides, measurement-noise sequences, reference, repeated-load schedule, initial state and 1 ms RK4 integration. The measured values themselves differ with each policy's plant trajectory. These are the new joint verification outcomes; the earlier 4900-series outcomes are identified as previous-design follow-up. The new correction contains 27 development paths and 164 full-horizon paths: 68 corrected SMC follow-ups and 96 new four-policy joint paths. Selection files, actual execution arguments, sampled inputs and every failed path accompany the checkpoints. The intervals are numerical design choices without physical-frequency interpretation.

Evidence versions. The earlier 255-path extension is preserved in its own result directory, together with the unmodified runner. That runner passed the 8 s argument only to PID; its active-SMC paths had zero baseline dwell and ungated discharge. Earlier claims that those SMC paths matched the latch/gate are withdrawn. The corrected dispatcher records the actual arguments alongside each checkpoint. The hydraulic/controller source is unchanged; the correction is in experiment dispatch. A configuration flag named matched is not itself evidence that the function received the requested arguments.

Table 14: Roles of corrected paths. Development precedes selection; the new joint vectors are frozen before corrected development. The original 158-path and earlier 255-path ledgers remain historical evidence.

Family	Seeds	Duration (s)	Paths	Role
SMC development	4600–4602	300	27	Selection only
SMC common nominal	4700–4709	1800	20	Known-protocol follow-up; two steps
SMC known vectors	4800–4823	1800	24	Known-design follow-up
SMC previous joint	4900–4923	1800	24	Previous-design follow-up
Four-policy new joint	5100–5123	1800	96	New stratified verification

Position-filter bias and moment bounds. The implemented position/velocity filter has the exact stored-state recursion

$$z_{k+1} = Fz_k + By_k, \qquad F = \begin{bmatrix} 1-\alpha & h \\ -\beta/h & 1 \end{bmatrix}, \quad B = \begin{bmatrix} \alpha \\ \beta/h \end{bmatrix}, \quad (h, \alpha, \beta) = (0.01, 0.08, 0.0005), \quad (1)$$

where $z = (\hat{x}, \hat{v})^\top$ and y is measured position. Its poles are 0.926834 and 0.993166. Bounded measurement inputs therefore give bounded filter states, but the DC response to a constant measured position is $(I - F)^{-1}B = (1, 0)^\top$: constant position bias is retained. For the additive iid measurement-noise component with a frozen input trajectory, $P = FPF^\top + \sigma_x^2 BB^\top$ gives position standard deviation $0.211675\sigma_x$ (0.318 mm at the nominal 1.5 mm raw SD). This is a filter-component calculation, not the covariance of the feedback-connected hydraulic loop. The identity $x_r - x = (x_r - \hat{x}) + (\hat{x} - x)$ shows why estimated error alone cannot certify true accuracy without an observer-error bound. The 2 mm design reserve accounts heuristically for one bias component; motion transients, pressure/load estimation and unbounded Gaussian samples prevent interpreting it as a guaranteed total error budget.

A limited moment bound is still available under the declared projections. Since true position is in $[0, L]$, bounded bias and finite measurement-noise variance give a uniformly finite L_2 norm of y_k. After expressing stored states and measurements in fixed dimensionless units, Schur stability supplies $C < \infty$ and $0 < r < 1$ such that $\|F^j\| \le Cr^j$. Minkowski's inequality then yields

$$\|z_k\|_{L_2} \le Cr^k \|z_0\|_{L_2} + \frac{C\|B\|}{1-r} \sup_i \|y_i\|_{L_2}. \quad (2)$$

This proves bounded second moments of this filter under projected-position inputs, including finite-variance Gaussian noise; it does not require feedback-input independence. The pressure filters have pole 0.82, and effective-load smoothing has pole 0.995 with input formed from those filters and a finite difference of $\hat{v}$, so their second moments are also bounded under bounded projected pressures and finite pressure-noise variance. Clipped control outputs and integrators remain bounded separately. These moment bounds establish neither a deterministic sample-path accuracy bound nor asymptotic tracking, invariance of the unprojected plant, or nonlinear switched stability.

Table 15: Corrected active-SMC development, nine candidates and three paths per candidate. Failures are ranked first, then mean J and candidate ID. Oil is mL; drop and RMSE are mm.

ID	g_t/g_a	Fail	Mean J	Drop	RMSE	Oil	Changes
0	0.7/0.5	0	0.4401	2.678	0.684	36.398	7.000
1	0.7/1.0	0	0.4780	2.564	0.670	61.757	7.000
2	0.7/1.5	0	0.5105	2.509	0.659	81.302	7.000
3	1.0/0.5	0	0.3648	2.325	0.545	23.324	7.000
4	1.0/1.0	0	0.3938	2.260	0.537	41.515	7.000
5	1.0/1.5	0	0.4096	2.213	0.532	53.419	6.333
6	1.3/0.5	0	0.3038	1.991	0.479	14.309	6.333
7	1.3/1.0	0	0.3242	1.954	0.476	26.523	6.333
8	1.3/1.5	0	0.3451	1.951	0.473	37.256	6.333

Across 81 development-only weight profiles, corrected SMC candidate frequencies are 6: 81. These are descriptive counts, not probabilities. Under the 8 mm development reserve its selected candidate is 6. The original hybrid/PID weight profiles remain in the earlier ledger and do not change their gains.

Table 16: Corrected SMC paired nominal steps; measurement/control updates remain at 10 ms. Means are shown except worst error.

Step (ms)	Pass	Drop (mm)	RMSE (mm)	Oil (mL)	Changes	Worst error (mm)
1.0	10/10	2.037	0.252	13.856	5.800	5.866
0.5	10/10	1.992	0.251	13.620	5.600	5.858

The archived unlatched nominal SMC at 0.5 ms had means 1.901 mm drop, 0.248 mm RMSE, 45.54 mL discharge and 36.6 changes. Those values concern its old (1.3,1.5) selection. The corrected candidate is (1.3,0.5); execution correction and reselection are not a one-factor causal comparison.

Table 17: Paired other-policy minus corrected-SMC differences on common nominal noise seeds. Positive values favor corrected SMC for that objective. Student-t intervals are descriptive model-design summaries, without physical-population or simultaneous-inference meaning.

Other policy	Quantity	Mean difference	Descriptive 95% interval
Scalar hybrid	Drop (mm)	0.116	[-0.213, 0.446]
Scalar hybrid	RMSE (mm)	1.022	[1.009, 1.035]
Scalar hybrid	Oil (mL)	10.171	[5.038, 15.303]
Scalar hybrid	Changes	48.600	[47.577, 49.623]
Active PID	Drop (mm)	1.563	[1.335, 1.791]
Active PID	RMSE (mm)	0.062	[0.059, 0.066]
Active PID	Oil (mL)	21.666	[19.272, 24.059]
Active PID	Changes	1.400	[0.709, 2.091]
Margin hybrid	Drop (mm)	0.033	[-0.274, 0.341]
Margin hybrid	RMSE (mm)	0.953	[0.933, 0.973]
Margin hybrid	Oil (mL)	139.534	[128.721, 150.347]
Margin hybrid	Changes	48.400	[46.641, 50.159]

Table 18: Paired other-policy minus corrected-SMC differences on new stratified joint design. Positive values favor corrected SMC for that objective. Student-t intervals are descriptive model-design summaries, without physical-population or simultaneous-inference meaning.

Other policy	Quantity	Mean difference	Descriptive 95% interval
Scalar hybrid	Drop (mm)	0.377	[0.069, 0.686]
Scalar hybrid	RMSE (mm)	0.325	[0.097, 0.554]
Scalar hybrid	Oil (mL)	161.305	[116.449, 206.161]
Scalar hybrid	Changes	67.792	[54.756, 80.827]
Active PID	Drop (mm)	1.828	[1.697, 1.959]
Active PID	RMSE (mm)	0.021	[0.007, 0.034]
Active PID	Oil (mL)	20.324	[17.849, 22.799]
Active PID	Changes	1.250	[0.832, 1.668]
Margin hybrid	Drop (mm)	0.093	[-0.127, 0.313]
Margin hybrid	RMSE (mm)	0.220	[0.025, 0.416]
Margin hybrid	Oil (mL)	231.548	[207.015, 256.081]
Margin hybrid	Changes	82.417	[67.240, 97.594]

Table 19: Every new joint maximum error in mm. Screen order is scalar hybrid / PID / corrected SMC / margin hybrid; P requires all declared criteria. Full vectors and metrics remain in checkpoints.

Seed	Scalar hybrid	Active PID	Active SMC	Margin hybrid	Screens
5100	9.068	6.211	6.530	7.684	P/P/P/P
5101	8.467	5.498	5.810	6.977	P/P/P/P
5102	8.670	5.765	6.081	7.242	P/P/P/P
5103	9.277	6.181	6.498	7.658	P/P/P/P
5104	9.463	6.490	6.865	7.957	P/P/P/P
5105	7.565	4.589	4.911	6.067	P/P/P/P
5106	8.075	5.169	5.488	6.647	P/P/P/P
5107	9.850	7.068	7.377	8.544	P/P/P/P
5108	9.754	6.881	7.203	8.355	P/P/P/P
5109	9.972	7.059	7.310	8.540	P/P/P/P
5110	9.872	6.830	7.155	8.310	P/P/P/P
5111	9.353	6.209	6.512	7.679	P/P/P/P
5112	6.337	4.155	3.734	4.887	P/P/P/P
5113	7.371	4.445	4.746	5.923	P/P/P/P
5114	7.562	4.556	4.875	6.023	P/P/P/P
5115	7.405	4.453	4.762	5.919	P/P/P/P
5116	7.184	4.272	4.592	5.740	P/P/P/P
5117	6.555	4.143	3.839	5.002	P/P/P/P
5118	7.975	4.838	5.159	6.318	P/P/P/P
5119	8.501	5.462	5.783	6.952	P/P/P/P
5120	6.160	3.982	3.576	4.627	P/P/P/P
5121	6.144	4.329	3.717	4.876	P/P/P/P
5122	8.512	5.303	5.612	6.751	P/P/P/P
5123	8.650	5.835	6.157	7.303	P/P/P/P

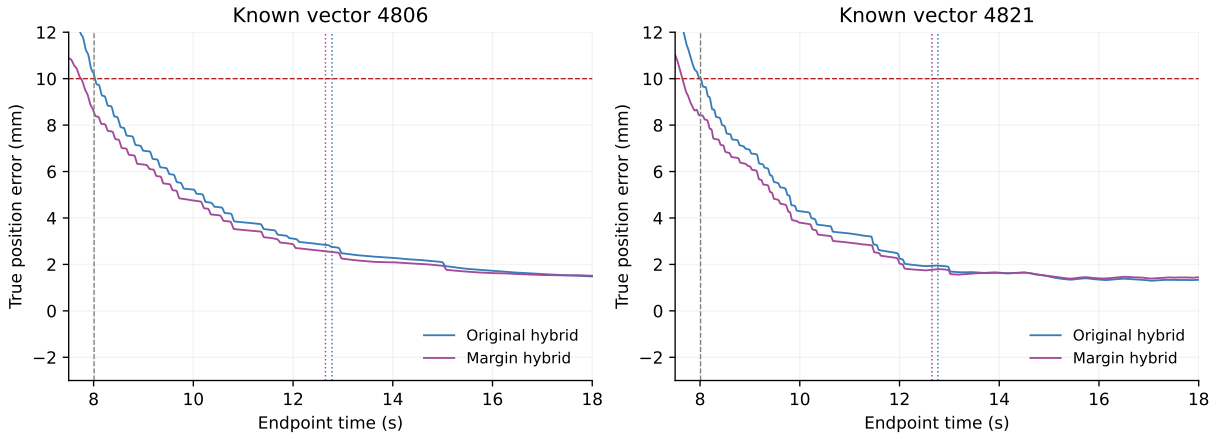


Figure 3: Short diagnostic replay of the two historical failures, keeping the 1800 s temperature slope. These known-case traces locate pre-hold transition errors and are not additional successful full-horizon paths.

```
python3 tools/run_paper1_matched_smc_correction.py --workers 8
python3 tools/report_paper1_matched_smc_correction.py
```

S12. Supply availability and boundary-resource accounting

The 88-path protocol and source hashes were frozen before their outcomes. Seeds 5200–5202 are exploratory development. Those explorations use different schedules and control prototypes and do not form a paired estimate of a return-path effect. The prospective paired storage contrast uses one common supply-only flow law. The archived double-port 88-path plan was not executed and contributes no outcomes or pass count. New normal seeds are 5500–5503 and new joint seeds are 5400–5407. Gains, the availability rule and the 1/0.5 ms refinement vector are fixed. No policy is selected by this new verification.

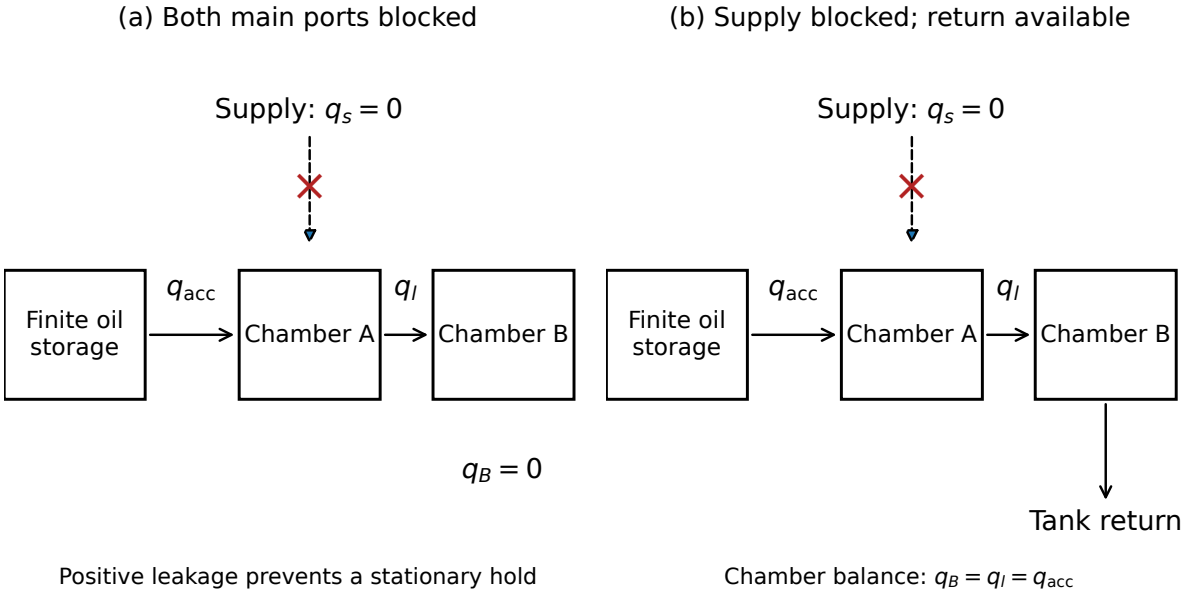


Figure 4: Declared hydraulic availability boundaries. Both-port blocking prevents the steady return required by positive internal leakage. Supply disconnection leaves the shared main-valve return orifice available. These are ideal virtual interventions; valve lag and finite gas storage remain in the model.

Table 20: All prospectively fixed joint-input intervals; SI base units unless the field name states degrees Celsius or a dimensionless scale.

Class	Input	Low	High
config	leakage_scale	3	8
config	accumulator_flow_scale	0.8	1.2
config	position_bias_m	-0.001	0.001
config	pressure_bias_pa	-50000	50000
config	pressure_noise_scale	1	1.5
config	temperature_end_c	40	55
config	post_step_load_n	12000	16000
plant	mass_kg	1000	1400
plant	stiffness_n_m	15000	21000
plant	viscous_n_s_m	1800	2600
plant	coulomb_n	680	1020
plant	stribeck_n	280	420
plant	effective_bulk_modulus_pa	1.6e+08	2.4e+08

The following paired intervals are t-based descriptive summaries of sampled differences. The joint design stratifies 13 inputs and is not an independently drawn physical population; its intervals

have no hardware-coverage or reliability interpretation. Normal-supply intervals concern four seeded noise paths at one theoretical plant. No p-values or physical pass probabilities are inferred.

Table 21: Selected coordinates of all eight joint vectors. Every remaining coordinate and exact floating-point value is in protocol.json; no vector is dropped.

Seed	Leak scale	x bias (mm)	p bias (kPa)	Load (kN)	Mass (kg)	β_e (10^8 Pa)
5400	5.592	-0.258	-28.82	15.335	1264.9	1.685
5401	5.365	-0.740	18.46	12.223	1386.5	2.284
5402	6.505	0.743	31.15	13.416	1239.3	2.029
5403	3.432	0.181	-12.59	15.696	1010.2	1.872
5404	7.467	0.805	44.33	14.582	1104.4	1.744
5405	6.984	-0.176	7.86	12.613	1313.4	2.189
5406	4.029	-0.983	-39.15	14.414	1084.2	1.988
5407	4.779	0.370	-3.28	13.617	1180.7	2.301

Table 22: Every paired interruption outcome. Pass column is enabled/disabled (1 means pass). Error and drop are mm; oil is enabled-storage mL. Every failure remains in the calculation.

Seed	Policy	Pass on/off	Error on	Error off	Drop on	Oil on
5400	Scalar hybrid	1/0	7.811	17.851	2.551	205.1
5401	Scalar hybrid	1/0	7.752	21.292	2.734	280.5
5402	Scalar hybrid	1/0	9.215	32.920	3.154	375.1
5403	Scalar hybrid	1/1	8.142	8.142	2.594	0.0
5404	Scalar hybrid	1/0	9.070	45.789	2.939	549.0
5405	Scalar hybrid	1/0	8.116	39.545	3.356	464.2
5406	Scalar hybrid	1/0	7.453	9.964	2.534	131.3
5407	Scalar hybrid	1/0	9.089	25.507	2.968	288.4
5400	Active PID	1/0	4.966	19.618	3.515	241.1
5401	Active PID	1/0	4.865	23.346	2.232	282.2
5402	Active PID	1/0	6.241	32.731	2.608	387.3
5403	Active PID	1/1	5.244	7.981	3.997	94.2
5404	Active PID	1/0	6.169	45.680	3.088	547.8
5405	Active PID	1/0	5.052	39.967	2.832	481.8
5406	Active PID	1/0	4.450	12.979	3.108	162.4
5407	Active PID	1/0	6.127	28.285	2.919	336.5
5400	Active SMC	1/0	5.268	45.867	2.186	564.5
5401	Active SMC	1/0	5.185	42.406	1.096	519.2
5402	Active SMC	1/0	6.567	55.041	1.462	663.0
5403	Active SMC	1/0	5.560	31.679	2.471	377.6
5404	Active SMC	1/0	6.487	68.079	1.876	832.1
5405	Active SMC	1/0	5.344	58.241	1.771	715.1
5406	Active SMC	1/0	4.770	35.429	1.736	437.7
5407	Active SMC	1/0	6.448	45.585	1.638	549.3
5400	Availability hybrid	1/0	7.811	17.712	1.978	345.6
5401	Availability hybrid	1/0	7.752	21.169	1.855	355.9
5402	Availability hybrid	1/0	9.215	32.795	1.735	455.4
5403	Availability hybrid	1/1	8.142	8.142	1.867	245.4
5404	Availability hybrid	1/0	9.070	45.658	2.000	642.3
5405	Availability hybrid	1/0	8.116	39.453	2.051	537.7
5406	Availability hybrid	1/0	7.453	9.879	1.757	268.4
5407	Availability hybrid	1/0	9.089	25.437	1.971	427.2

Table 23: Complete-path storage attribution. Paired changes use the same plant, noise and return path. The on/off intervention alone changes storage flow.

Policy	On pass/off fail seeds	On fail/off pass seeds	Both pass	Both fail
Scalar hybrid	5400,5401,5402,5404,5405, 5406,5407	None	1	0
Active PID	5400,5401,5402,5404,5405, 5406,5407	None	1	0
Active SMC	5400,5401,5402,5403,5404, 5405,5406,5407	None	0	0
Availability hybrid	5400,5401,5402,5404,5405, 5406,5407	None	1	0

Table 24: Normal paired scalar-hybrid minus active-SMC differences. Units follow the metric field; negative values indicate a smaller hybrid quantity.

Metric	Mean	Lower	Upper
hold_drop_mm	0.3692	-0.3315	1.07
post_transition_rmse_mm	1.019	1.008	1.031
post_transition_max_abs_error_mm	2.606	2.515	2.696
post_transition_main_supply_volume_ml	-1.232e+04	-1.246e+04	-1.219e+04
accumulator_oil_used_ml	8.056	-2.098	18.21
post_transition_storage_boundary_work_j	111.8	-28.98	252.5
post_transition_boundary_resource_j	-2.586e+05	-2.615e+05	-2.558e+05
isolated_max_abs_error_mm	0	0	0

Table 25: Storage-enabled joint family: scalar hybrid minus active SMC. All eight pairs are included; units follow field names. Intervals are descriptive.

Metric	Mean	Lower	Upper
hold_drop_mm	1.074	0.5733	1.575
post_transition_rmse_mm	0.5502	0.3916	0.7087
post_transition_max_abs_error_mm	2.627	2.565	2.69
post_transition_main_supply_volume_ml	-4364	-4546	-4181
accumulator_oil_used_ml	-295.6	-337.5	-253.8
post_transition_storage_boundary_work_j	-3454	-4095	-2812
post_transition_boundary_resource_j	-9.509e+04	-9.935e+04	-9.083e+04
isolated_max_abs_error_mm	1.892	1.556	2.228

Table 26: Storage-enabled joint family: availability hybrid minus scalar hybrid. All eight pairs are included; units follow field names. Intervals are descriptive.

Metric	Mean	Lower	Upper
hold_drop_mm	-0.9521	-1.192	-0.7123
post_transition_rmse_mm	-0.6262	-0.9472	-0.3051
post_transition_max_abs_error_mm	0	0	0
post_transition_main_supply_volume_ml	-53.95	-73.14	-34.77
accumulator_oil_used_ml	123.1	74.97	171.1
post_transition_storage_boundary_work_j	1510	836.4	2184

Metric	Mean	Lower	Upper
post_transition_boundary_resource_j	377.4	-142.5	897.3
isolated_max_abs_error_mm	-2.398	-2.986	-1.809

Table 27: Prespecified seed 5400: fine minus coarse. Steps are 0.5 and 1 ms. Pass is coarse/fine; error and drop differences are mm; supply and oil differences are mL.

Policy	Store	Pass	Drop	Error	Supply	Oil
Scalar hybrid	on	1/1	-0.1996	-0.0010	4.661	-4.730
Scalar hybrid	off	0/0	-0.1554	-0.0412	-1.914	0.000
Active PID	on	1/1	-0.1892	-0.0175	21.945	-0.117
Active PID	off	0/0	-0.2120	-0.0361	25.254	0.000
Active SMC	on	1/1	-0.0371	-0.0042	4.646	0.976
Active SMC	off	0/0	-0.1954	-0.1950	-0.171	0.000
Availability hybrid	on	1/1	-0.1580	-0.0010	-2.172	1.444
Availability hybrid	off	0/0	-0.1496	-0.0354	-2.169	0.000

Table 28: All failed seeds by family and policy. Every failure is included in its summary.

Family	Policy	Failed seeds
normal	Scalar hybrid	None
normal	Active PID	None
normal	Active SMC	None
normal	Availability hybrid	None
isolated on	Scalar hybrid	None
isolated on	Active PID	None
isolated on	Active SMC	None
isolated on	Availability hybrid	None
isolated off	Scalar hybrid	5400, 5401, 5402, 5404, 5405, 5406, 5407
isolated off	Active PID	5400, 5401, 5402, 5404, 5405, 5406, 5407
isolated off	Active SMC	5400, 5401, 5402, 5403, 5404, 5405, 5406, 5407
isolated off	Availability hybrid	5400, 5401, 5402, 5404, 5405, 5406, 5407
refined on	Scalar hybrid	None
refined on	Active PID	None
refined on	Active SMC	None
refined on	Availability hybrid	None
refined off	Scalar hybrid	5400
refined off	Active PID	5400
refined off	Active SMC	5400
refined off	Availability hybrid	5400

Full normal, interruption, refinement, paired and conservation data are supplied in the reproduction archive. Differences involving failed paths remain included.

```
python3 tools/run_paper1_supply_validation.py --workers 8
python3 tools/report_paper1_supply_coordination.py
python3 tools/check_paper1_supply_coordination.py --preflight
```

S13. Equal-budget resource-oriented retuning

The separate protocol fixes nine candidates and three complete 1800 s development paths per policy before any new outcome. Both policies use $g_t \in \{0.7, 1.0, 1.3\}$ and $g_a \in \{0.3, 0.5, 0.7\}$; candidate IDs follow that Cartesian-product order. Development seeds are 5600–5602. The repeated loads are $(0, 9), (60, 14), (300, 10), (600, 16), (900, 11), (1200, 15)$ in s/kN, with fivefold leakage, normal supply, 10 ms sensing/control and 1 ms coupled RK4. There is no retuning of the plant or observer. Failed paths rank before B only in the sense that fewer failures always take priority; zero failures are required for a development-feasible candidate. Mean B then ranks candidates, with ID breaking an exact tie. B uses the common 8.01–1800 s window. Both policies retain the same main-valve limit and explicit 8 s latch/discharge gate. Other architecture parameters, including SMC amplitude/width and hybrid thresholds, remain fixed. This bounded search does not establish a globally optimized control-family frontier.

Selection is written to a hash-bound ledger before the independent validation tasks are executed. Validation seeds 5620–5623 are disjoint from development and all earlier normal-supply seeds. The experiment adds 54 development and eight validation paths; it does not overwrite the earlier 88-path supply study. Its normal-supply validation does not extend the interruption or joint-uncertainty evidence. Machine-readable protocol, complete candidate records, selection, validation protocol and all 62 checkpoints accompany the reproduction ZIP.

Table 29: All development candidates; three paths each. Failure counts use the full feasibility screen.

Policy	ID	g_t	g_a	Fail	Mean B (kJ)	RMSE (mm)
Hybrid	g00	0.7	0.3	3	23.395	1.314
Hybrid	g01	0.7	0.5	3	23.399	1.315
Hybrid	g02	0.7	0.7	3	23.440	1.315
Hybrid	g03	1.0	0.3	0	22.760	1.243
Hybrid	g04	1.0	0.5	0	22.862	1.243
Hybrid	g05	1.0	0.7	0	22.924	1.245
Hybrid	g06	1.3	0.3	0	21.832	1.202
Hybrid	g07	1.3	0.5	0	22.251	1.185
Hybrid	g08	1.3	0.7	0	22.251	1.183
SMC	g00	0.7	0.3	0	211.695	0.304
SMC	g01	0.7	0.5	0	211.550	0.302
SMC	g02	0.7	0.7	0	211.632	0.300
SMC	g03	1.0	0.3	0	247.442	0.261
SMC	g04	1.0	0.5	0	247.334	0.260
SMC	g05	1.0	0.7	0	247.273	0.259
SMC	g06	1.3	0.3	0	285.747	0.248
SMC	g07	1.3	0.5	0	285.576	0.247
SMC	g08	1.3	0.7	0	285.671	0.247

Table 30: Every selected-policy validation path. Error is the maximum post-transition absolute error; drop uses confirmed hold.

Policy	Seed	Pass	Error (mm)	Drop (mm)	RMSE (mm)	B (kJ)
Hybrid	5620	1	6.566	2.674	1.197	22.606
Hybrid	5621	1	6.917	2.408	1.204	22.541
Hybrid	5622	1	7.022	2.626	1.224	22.634
Hybrid	5623	1	6.413	2.215	1.207	22.079
SMC	5620	1	9.391	2.673	0.309	206.558
SMC	5621	1	9.809	2.930	0.308	211.537

Policy	Seed	Pass	Error (mm)	Drop (mm)	RMSE (mm)	B (kJ)
SMC	5622	1	9.913	2.579	0.309	208.097
SMC	5623	1	9.350	2.835	0.308	211.070

Table 31: Paired hybrid minus SMC validation differences, $n = 4$. Intervals use a Student t multiplier with three degrees of freedom; they describe seeded simulation variability only.

Metric	Mean	95% interval
Boundary resource (kJ)	-186.850	[-190.908, -182.793]
Main-source volume (mL)	-8941.490	[-9127.368, -8755.611]
Post-transition RMSE (mm)	0.900	[0.882, 0.918]
Confirmed-hold drop (mm)	-0.274	[-0.825, 0.277]
Stored-oil discharge (mL)	68.041	[43.748, 92.334]
Mode changes	48.000	[48.000, 48.000]

All path metrics, including storage-outlet work, oil, mode changes, pressure and boundary counters, remain in the JSON checkpoints. Candidate ranking is independent for each policy and does not equalize RMSE; lower B with larger tracking error is a resource–accuracy trade-off. Four validation seeds are a small nominal screen. No finite grid certifies an optimum, physical reliability, robustness over an uncertainty region or actual energy saving.